

AHMED ISLAM

AI / Agentic AI Engineer | LLM Systems | Python & Data
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PROFESSIONAL SUMMARY

Computer Science graduate with a year of production experience building LLM-powered agents and multi-agent systems with LangGraph, LangChain, and the Model Context Protocol. I design tool-calling agents, engineer and version system prompts, and handle agent memory using vector stores alongside Redis and PostgreSQL. I also work across the wider stack, from Python ETL pipelines and SQL to Docker, AWS, and Azure. Currently building AI infrastructure at Padelos.

EXPERIENCE

Associate AI Engineer | Padelos

Jan 2026 – Present

- Build LLM-powered agents and multi-agent workflows on the Padelos SaaS platform with LangChain and LangGraph, automating multi-step production processes.
- Developed a Model Context Protocol (MCP) server that exposes platform data and business logic to AI agents through structured tool-calling interfaces.
- Engineer and version system prompts for PadelGPT, the platform assistant, and built a quality-evaluation pipeline that grades responses against a rubric and generates structured Excel and CSV reports.
- Manage agent memory with vector stores (FAISS, ChromaDB) for retrieval and Redis and PostgreSQL for short-term and persistent state.
- Built n8n pipelines that extract, transform, and report conversation data from Azure Cosmos DB across player and admin schemas.
- Migrated the platform chatbot from Azure to AWS EC2 and shipped services with Docker and Bitbucket Pipelines CI/CD, adding CloudWatch-based observability.

Junior AI Engineer | IR Solutions

Jul – Dec 2025

- Built production agentic AI systems with n8n and engineered large-scale web scraping and data-ingestion pipelines.
- Delivered end-to-end AI agents from design through deployment in a fast-paced team.

Deep Learning Intern | Itsolera PVT LTD

Jul – Oct 2024

- Built a CNN content-moderation system for images and text, plus a facial-recognition attendance system.
- Implemented real-time product-detection models for retail shelf monitoring with computer vision.

Machine Learning Fellow | Bytewise LTD

Jun – Sep 2024

- Completed an intensive fellowship in supervised and unsupervised ML, feature engineering, and model evaluation.
- Applied end-to-end ML workflows on real datasets with scikit-learn and pandas.

PROJECTS

Multimodal RAG System

- Built a RAG system combining text and image understanding with LangChain, FAISS/ChromaDB, and Hugging Face Transformers.
- FastAPI backend serving quiz generation, Q&A, and mock-interview simulations from uploaded documents and images.

Open Source Contributor | modelcontextprotocol/inspector

- Fixed a rendering bug where nested and optional array parameters fell back to raw JSON instead of structured form inputs.
- Resolved styling inconsistencies across tabs, improving the developer experience for the wider MCP ecosystem.

Brain Tumor Segmentation | U-Net

- Trained a 2D U-Net on the BraTS2020 MRI dataset to segment necrotic core, edema, and enhancing tumor regions.
- Reached 99.2% validation accuracy and a 0.71 Dice score using custom augmentation to handle class imbalance.

Customer Churn Prediction | Streamlit Dashboard

- Cleaned and preprocessed data, handled class imbalance, and compared multiple classifiers to improve accuracy.
- Built an interactive Streamlit dashboard with filters and multi-chart visualizations.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C++

LLM & Agentic AI: LangChain, LangGraph, MCP, RAG, tool calling, multi-agent orchestration, prompt engineering, OpenAI, Anthropic Claude, Google Gemini, LLaMA, Mistral, Ollama, Hugging Face, Groq API

Memory & Vector Stores: FAISS, ChromaDB, Redis, PostgreSQL

Data & Backend: FastAPI, Flask, FastMCP, ETL/ELT pipelines, n8n, REST APIs, pandas

Databases: PostgreSQL, MySQL, SQLite, MongoDB

ML & DL: scikit-learn, TensorFlow, PyTorch, CNNs, U-Net, YOLO, NLP, Computer Vision, Transfer Learning

Cloud & DevOps: AWS (EC2, CloudWatch), Azure (App Service), Docker, Kubernetes, Bitbucket Pipelines (CI/CD), Git

Visualization: Power BI, Streamlit, Gradio

EDUCATION

BS Computer Science | University of Haripur | CGPA 3.79/4.0

Sep 2021 – Jul 2025

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Computer Vision, Database Systems, Software Engineering.

CERTIFICATIONS

- Data Science & AI Bootcamp, Atomcamp
- IBM Certified Machine Learning
- AI & Machine Learning Foundation, Xeven Solutions
- MLOps Course, Udemy
- Introduction to LangGraph, LangChain Academy